SANGBAY
Gas Sensor

Hydrogen Sulfide Sensor

S4H_2S Sensor Specification

Electrochemical Hydrogen Sulfide Sensor

Small size | Low consumption | Long life | Low cost | High precision

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1. Product Overview

S4H_2S gas detection sensor is a kind of 3 electrodes electrochemical sensor. Hydrogen sulfide is oxidized on working electrode and oxygen is reduced on the counter electrode. The reference electrode contacts with the electrolyte to keep working electrode at a stable potential. The redox reaction current is directly proportional to the concentration of H_2S.



2. Application Field

(1) H_2S alarm for both portable and fixed gas detection devices.

(2) Industrial sites hydrogen sulfide gas detection.

(3) H_2S emission in all industry.

(4) Environment air quality detection.

3. Product Characteristics

(1) High precision and low cost.

(2) Widely used in all kinds of H_2S gas detection industry.

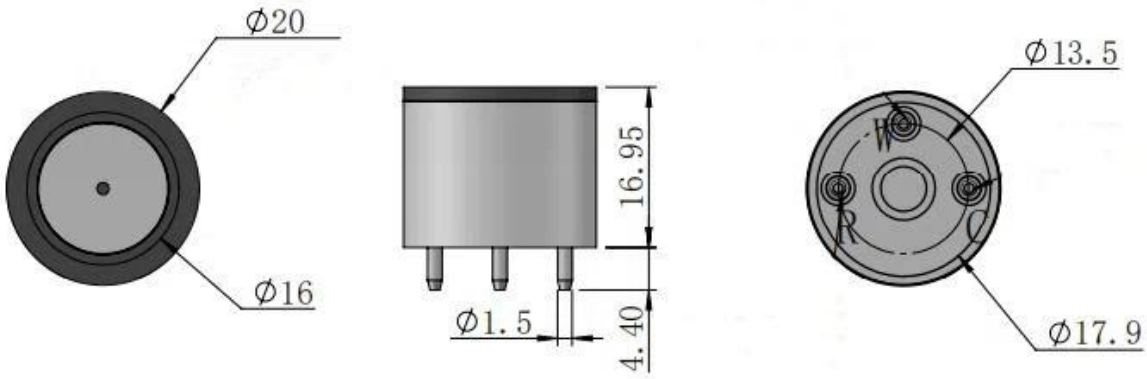
(3) Fast response time and recovery.

(4) Robust environment performance.

(5) Good repeatability and stability.

4. Technical Parameters

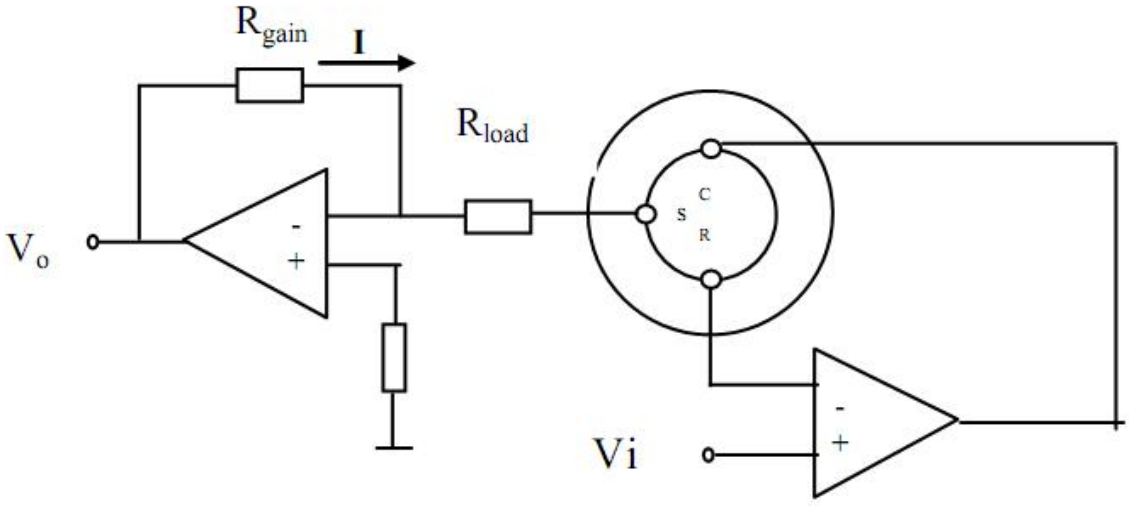
Description	Specification
Product Name	S4H_2S-100
Technology	Electrochemical Three electrodes
Measurement Gas	H_2S
Measurement Range	0-100 ppm
Maximum Overload	500 ppm
Output Signal	650±200 nA/ppm
Zero Drift	±1 ppm
Resolution	1 ppm
Response Time	< 30 s
Bias Voltage	0
Recommended Load Resistor	5~30 Ω
Temperature Range	-30~50 ℃
Humidity Range	15%~ 90%RH (no condensation)
Repeatability	< ±2% output signal
Pressure Range	90 ~ 110 kPa
Long-term Stability	< 2% signal/month
Linearity	Linear
Shelf life	12 months after delivery
Service life	2 years



Product Dimensions in mm

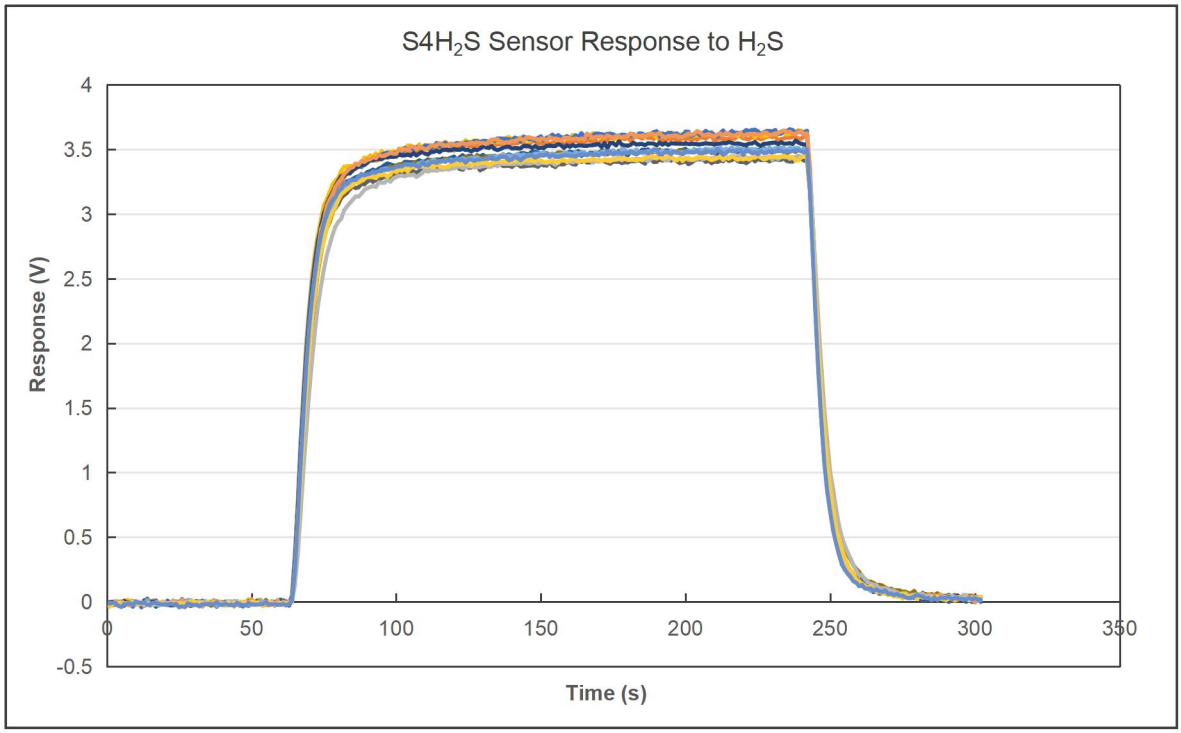
All tolerances ±0.20mm

5. Basic Circuit

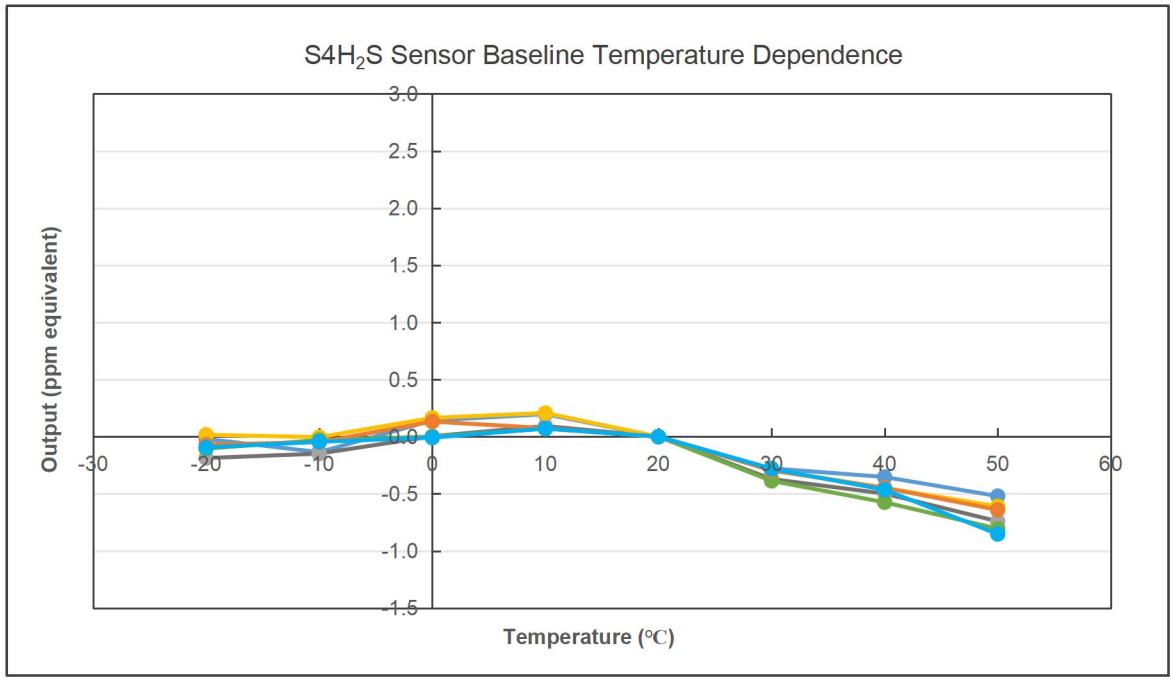


Note:Above picture shows the basic test circuit of the S4H_2S sensor.

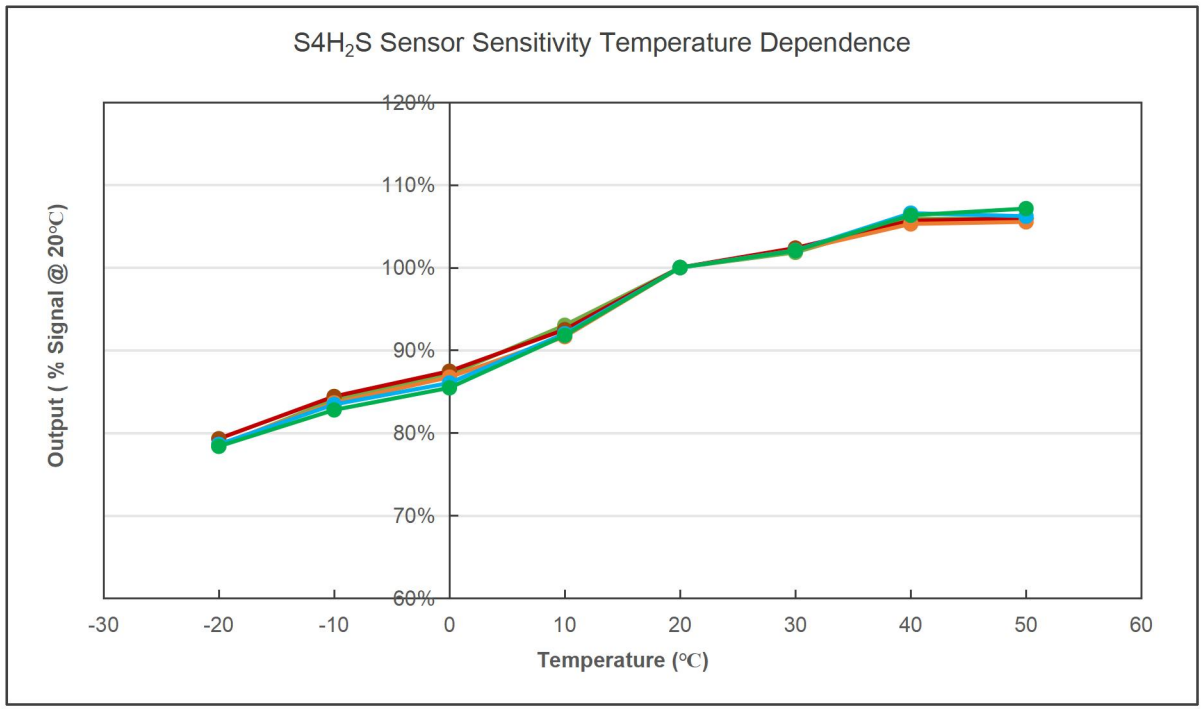
6. Sensor Characteristic Description



S4H_2S Sensor Response to H_2S gas



S4H_2S sensor baseline temperature dependence



S4H_2S sensor sensitivity temperature dependence

7. Cross Sensitivity

Interfering gas	Concentration （ppm）	Displayed（ppm H_2S）
CO	300	1
SO_2	20	1
NH_3	50	0
NO_2	20	-1
H_2	1000	0
NO	50	2
Cl_2	20	-1.4